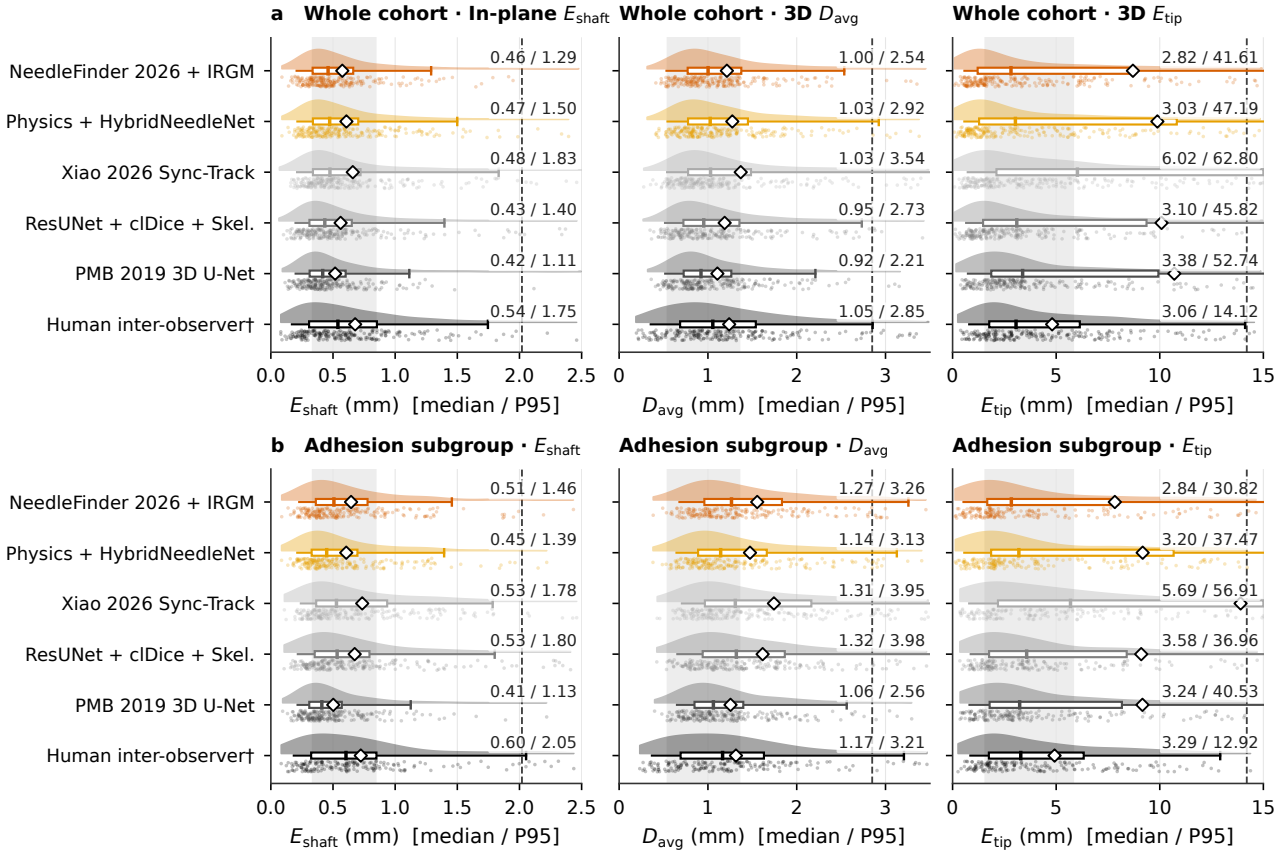

a Whole cohort · In-plane E_{shaft}
Whole cohort · 3D D_{avg}
Whole cohort · 3D E_{tip}
NeedleFinder 2026 + IRGM
Physics + HybridNeedleNet
Xiao 2026 Sync-Track
ResUNet + clDice + Skel.
PMB 2019 3D U-Net
Human inter-observer†
0.46 / 1.29
0.47 / 1.50
0.48 / 1.83
0.43 / 1.40
0.42 / 1.11
0.54 / 1.75
1.00 / 2.54
1.03 / 2.92
1.03 / 3.54
0.95 / 2.73
0.92 / 2.21
1.05 / 2.85
2.82 / 41.61
3.03 / 47.19
6.02 / 62.80
3.10 / 45.82
3.38 / 52.74
3.06 / 14.12
0.0 0.5 1.0 1.5 2.0 2.5
0 1 2 3
0 5 10 15
E_{shaft} (mm) [median / P95]
D_{avg} (mm) [median / P95]
E_{tip} (mm) [median / P95]
b Adhesion subgroup · E_{shaft}
Adhesion subgroup · D_{avg}
Adhesion subgroup · E_{tip}
NeedleFinder 2026 + IRGM
Physics + HybridNeedleNet
Xiao 2026 Sync-Track
ResUNet + clDice + Skel.
PMB 2019 3D U-Net
Human inter-observer†
0.51 / 1.46
0.45 / 1.39
0.53 / 1.78
0.53 / 1.80
0.41 / 1.13
0.60 / 2.05
1.27 / 3.26
1.14 / 3.13
1.31 / 3.95
1.32 / 3.98
1.06 / 2.56
1.17 / 3.21
2.84 / 30.82
3.20 / 37.47
5.69 / 56.91
3.58 / 36.96
3.24 / 40.53
3.29 / 12.92
0.0 0.5 1.0 1.5 2.0 2.5
0 1 2 3
0 5 10 15
E_{shaft} (mm) [median / P95]
D_{avg} (mm) [median / P95]
E_{tip} (mm) [median / P95]